Joint state-of-charge and drifting current-sensor-bias estimation for lithium-ion batteries: an augmented-state equivalent-circuit observer evaluated against physics-based virtual cells

Author One[a,*], Author Two[a]

[a] Affiliation, Address

[*] Corresponding author: email@institution.edu

Abstract

[TODO: Write after results are finalized.] Accurate state-of-charge (SOC) estimation in battery management systems relies on current and voltage measurements whose accuracy degrades over the system lifetime. This work studies the joint estimation of SOC and a slowly *growing* current-measurement bias—originating from sensor ageing as well as from un-modeled cabling and interconnection resistances—using an equivalent-circuit-model (ECM) observer augmented with a bias state. The observer is evaluated against physics-based virtual cells implemented in PyBaMM with three published parameter sets (NMC, LCO and NMC/silicon–graphite chemistries), under constant-current, CC–CV and dynamic-stress-test (DST) operation, across temperature, C-rate, depth-of-discharge, drift-rate, sensor-age and voltage-bias variations. [TODO: Add quantitative highlights and the experimental-validation summary.]

1 Introduction

The state of charge (SOC) of a lithium-ion battery is not directly measurable and must be inferred from terminal quantities, most commonly by combining coulomb counting with voltage-based correction in a model-based observer—most commonly an equivalent-circuit model (ECM) [1] combined with an extended Kalman filter (EKF) or one of its variants [2–4]. The fidelity of this inference is bounded by the quality of the current and voltage measurements. While measurement *noise* is explicitly accommodated by the stochastic formulation of the filter, measurement *bias* is not: a systematic current offset is integrated by the coulomb-counting dynamics and produces an unbounded drift of the SOC estimate, while the voltage-based correction can only partially counteract it, and may even be misled in flat regions of the open-circuit voltage (OCV) curve. Comparative studies confirm that bias-type current errors, unlike zero-mean noise, accumulate through the current integral and can cause estimator divergence [5], and error-source surveys rank current-measurement offsets among the dominant contributors to online SOC estimation error [6, 7]. Lin [8] derived the SOC estimation bias and variance as explicit functions of the sensor noises, the battery parameters and the observer tuning, exposing a fundamental trade-off between suppressing the bias-induced and the variance-induced error components—a trade-off that an explicit bias state is designed to break.

Current-measurement bias in battery systems has several physical origins. Hall-effect and shunt-based sensors exhibit offset errors that vary with temperature and grow with ageing of the sensing element and its conditioning electronics; calibration errors introduce quasi-static offsets that may change abruptly after maintenance events. In addition, effects that are not sensor faults at all can manifest as an equivalent measurement bias at the observer level: un-modeled resistances of cabling, busbars, contact and weld interfaces re-route or offset the effective current seen by the model, and their slow growth with interconnect ageing produces a bias-like, time-varying disturbance. From the observer's perspective these contributions lump into a single,

slowly drifting disturbance term, which motivates estimating it online rather than assuming it constant or zero.

The idea of augmenting a battery state estimator with sensor-bias states is established. Zhao et al. [9] derived necessary and sufficient observability conditions for a nonlinear ECM using a differential-geometric approach, and demonstrated—including experimental validation on an NMC pouch cell—that current- and voltage-sensor biases can be estimated by a nonlinear Kalman filter acting on an augmented model; importantly, they also showed that the simultaneous estimation of both current- and voltage-sensor biases renders the SOC ambiguous, since the filter cannot attribute an output error to one sensor or the other. Yoo et al. [10] developed an EKF-based current-sensor bias estimator for a hybrid urban-air-mobility application and validated it with Hall-sensor measurements, maintaining SOC errors within 2 % under realistic flight profiles. Dual-filter formulations that decouple the bias estimate from the electrochemical states have recently been proposed for chemistries with flat OCV characteristics [11]. Beyond estimation, the fault-diagnosis literature treats sensor bias as one of several fault classes (bias, gain, drift, intermittent) to be detected and reconstructed by dedicated observers [12].

Three gaps remain. First, the bias is almost universally modeled and injected as a *constant* (or step) offset, whereas the degradation mechanisms above produce a bias that *grows in time*, on time scales comparable to or slower than a single operating cycle; the ability of an augmented observer to *track* such drift, and the excitation it requires, is not systematically characterized. Second, validation is typically performed on a single cell and chemistry; since the bias enters the measurement equation through the ohmic resistance and the OCV slope, its identifiability is strongly chemistry- and operating-point-dependent, which calls for a controlled multi-chemistry assessment. Third, synthetic studies usually corrupt data generated by the *same* model class used by the observer, hiding the interaction between model mismatch and bias estimation; a physics-based plant of a different model class provides a more honest testbed.

This paper addresses these gaps with the following contributions:

1. a simulation framework in which the plant is a physics-based single-particle model with electrolyte (SPMe) implemented in PyBaMM [13], instantiated with three published parameter sets—Mohtat2020 [14], Ai2020 [15] and OKane2022 [16] (based on the LG M50 parameterization of Chen et al. [17])—while the observer uses a deliberately mismatched second-order ECM;

2. a sensor-degradation model comprising an initial offset ("sensor age"), a slow drift in time, step faults and random-walk components on the current channel, and a constant or drifting bias on the voltage channel, with magnitudes anchored to published datasheet figures for automotive current sensing [18]; observers augmented with the current bias, the voltage bias, or both are compared, the last probing the ambiguity of [9] quantitatively;

3. a joint battery-aging / sensor-aging study in which aged virtual cells (loss of active material and ohmic resistance growth, down to 80 % SOH) are observed with the fresh-cell calibration, quantifying both the growing benefit of the bias state and the confounding between sensor diagnosis and aging;

4. a bias-augmented EKF on the ECM, compared one-to-one against its non-augmented counterpart over a full-factorial-inspired campaign spanning chemistry, load profile (constant current, CC–CV cycle, and a dynamic stress test scaled from the USABC procedure [19]), temperature, peak C-rate, depth of discharge, drift rate, sensor age and voltage bias;

5. a joint battery-ageing × sensor-ageing campaign in which the plant is aged (loss of active material and resistance growth) while the observer retains its beginning-of-life calibration, demonstrating that the value of the bias state *grows* with battery age: the estimated bias absorbs the lumped disturbance created by capacity fade and resistance growth in addition

to the physical sensor offset, keeping the SOC error bounded where the baseline observer degrades;

6. [TODO: experimental validation on a laboratory cell tester with software-injected, ground-truth-known drifting bias and a physical series resistance in the cabling.]

The remainder of the paper is organized as follows. Section 2 describes the virtual plant, the observer models, the sensor-degradation models and the design of the simulation campaign. Section 3 reports the results. Section 4 concludes.

2 Methodology

2.1 Physics-based virtual plant

The plant is the single-particle model with electrolyte dynamics (SPMe) [20]—a reduced-order descendant of the Doyle–Fuller–Newman model [21]—as implemented in PyBaMM v26 [13], solved isothermally at a prescribed ambient temperature T. Three published parameter sets are used, providing chemically distinct virtual cells: Mohtat2020 (5 Ah NMC/graphite pouch cell [14]), Ai2020 (2.28 Ah LCO/graphite pouch cell [15]) and OKane2022 (5 Ah NMC811 / silicon–graphite cylindrical cell [16, 17]). Using a physics-based plant of a different model class than the observer introduces a realistic, structured model mismatch: the residual voltage error of the calibrated observer model against the plant is 13 mV to 30 mV RMS depending on chemistry and temperature (Section 2.2), comparable to what is observed when fitting ECMs to laboratory data.

2.2 Observer model and calibration

The observer uses a second-order ECM. With the PyBaMM sign convention ($i > 0$ in discharge), the continuous-time model is

$$\dot{\mathrm{SOC}} = -\frac{i}{3600\,Q}, \qquad \dot{v}_j = -\frac{v_j}{\tau_j} + \frac{R_j}{\tau_j}\,i, \quad j = 1, 2, \tag{1}$$

$$v = U(\mathrm{SOC}) - R_0\,i - v_1 - v_2, \tag{2}$$

where Q is the usable capacity, $U(\cdot)$ the pseudo-OCV curve, R_0 the ohmic resistance and (R_j, τ_j) the RC polarization branches.

The pseudo-OCV curve is obtained from the plant by a C/25 discharge–charge pair between the voltage limits, taking the mean of the two branches on a common SOC axis. Averaging the branches removes both the small residual overpotential and the OCV hysteresis, which is significant for the silicon-containing negative electrode of the OKane2022 cell. The usable capacity Q is defined as the charge throughput of the C/25 discharge branch, and SOC is referenced to this window throughout.

The ECM parameters $\theta = (R_0, R_1, \tau_1, R_2, \tau_2)$ are calibrated by nonlinear least squares against *bias-free commissioning data*: a 1 C constant-current discharge of the plant at the operating temperature. The simulated ECM terminal voltage, driven by the true current and evaluated at the true SOC, is fitted to the plant voltage with a trust-region-reflective solver under box constraints, and samples below 5 % SOC are excluded to prevent the steep OCV tail from dominating the cost. This mimics the realistic workflow in which the model is calibrated when the pack is new and the instrumentation trusted, and subsequently deployed while the sensors degrade. The calibration is repeated for every (chemistry, temperature) pair used in the campaign.

2.3 Sensor degradation model

The observer receives corrupted measurements

$$i_m(t) = i(t) + b(t) + w_i(t), \qquad v_m(t) = v(t) + b_v(t) + w_v(t), \tag{3}$$

with zero-mean white noises $w_i \sim \mathcal{N}(0, \sigma_i^2)$, $w_v \sim \mathcal{N}(0, \sigma_v^2)$ ($\sigma_i = 20\,\mathrm{mA}$, $\sigma_v = 2\,\mathrm{mV}$). The current-sensor bias is modeled as

$$b(t) = b_0 + \dot{b}\,t + \Delta b \mathbb{1}(t \geq t_s) + \beta(t), \tag{4}$$

where b_0 is the offset accumulated before the test ("sensor age"), $\dot{b}$ a constant drift rate representing slow degradation of the sensor and of the interconnect resistances, Δb an optional step (calibration-type fault) at time t_s, and $\beta(t)$ an optional Wiener process. The magnitudes are chosen to be representative of automotive-grade instrumentation: open-loop Hall-effect pack-current sensors exhibit offset errors of the order of 0.5 % to 1 % of full scale, with a residual offset that drifts with temperature, package stress and ageing even after factory compensation [22]; for a 15 A to 25 A full-scale sensor this corresponds to $b_0 \approx 0.1\,\mathrm{A}$ to $0.25\,\mathrm{A}$, while the within-test drift— dominated by the thermal excursion of the sensor during operation—is of the order of tens of mA per hour. The baseline scenario therefore uses $b_0 = 0.1\,\mathrm{A}$ and $\dot{b} = 0.02\,\mathrm{A/h}$, with sweeps from $0.005\,\mathrm{A/h}$ to $0.05\,\mathrm{A/h}$ and a dedicated 0.1 A step-fault case. The voltage-sensor bias $b_v(t)$ is constant or slowly drifting.

The magnitudes are anchored to published figures for automotive sensing: Hall-effect and shunt current sensors exhibit lifetime offset and sensitivity drifts of the order of 0.1 % to 1 % of full scale [18], i.e. tens to a few hundred mA for a cell-level sensor, accumulating over months of operation; battery-monitoring analog front ends specify total voltage measurement errors of 1 mV to 2 mV. Within a single test the physical drift is therefore quasi-static and is represented by the initial offset b_0, which grows across life stages; scenarios with in-test drift rates $\dot{b} \geq 0.05\,\mathrm{A/h}$ are labeled *accelerated* and serve as stress tests of the drift-tracking bandwidth. Larger lumped offsets remain plausible when un-modeled cabling and interconnect resistances contribute, which is precisely the lumped-disturbance interpretation adopted here.

2.4 Battery aging emulation

To study the interaction between sensor degradation and battery degradation, aged virtual cells are generated from the fresh parameter sets by two mechanisms. Loss of active material (LAM) is emulated by scaling the active-material volume fractions of both electrodes by a factor equal to the target state of health (SOH), producing an approximately proportional capacity fade. Ohmic resistance growth—the lumped effect of SEI growth and of contact/interconnect degradation—is applied as an exact series resistance $\Delta R = 2\,R_0^{\mathrm{fresh}}\,(1 - \mathrm{SOH})$ on the terminal voltage. SOH levels of 100 %, 95 %, 90 %, 85 % and 80 % are considered. Crucially, the observer retains the *fresh* calibration (Q, ECM parameters and pseudo-OCV) at all SOH levels, reproducing the realistic situation of a BMS whose model has not been re-identified as the pack ages. The true SOC of the aged cell is referenced to the aged usable capacity. This design probes the lumped-disturbance interpretation of the bias state: capacity fade injects a coulombic error and resistance growth injects a voltage error, both of which the augmented filter can partially absorb into $\hat{b}$—improving SOC accuracy at the cost of biasing the sensor-fault diagnosis, a confounding that is quantified in Section 3.

2.5 Bias-augmented extended Kalman filter

Discretizing (1) exactly over a step Δt_k and augmenting the state with the bias, modeled as a random walk, gives $x_k = [\mathrm{SOC}_k,\, v_{1,k},\, v_{2,k},\, b_k]^\top$ with

$$\mathrm{SOC}_{k+1} = \mathrm{SOC}_k - \frac{(i_{m,k} - b_k)\,\Delta t_k}{3600\,Q}, \tag{5}$$

$$v_{j,k+1} = a_{j,k}\,v_{j,k} + R_j\,(1 - a_{j,k})\,(i_{m,k} - b_k), \qquad a_{j,k} = e^{-\Delta t_k/\tau_j}, \tag{6}$$

$$b_{k+1} = b_k + \eta_k, \qquad \eta_k \sim \mathcal{N}(0, q_b), \tag{7}$$

and measurement model

$$v_{m,k} = U(\mathrm{SOC}_k) - R_0\,(i_{m,k} - b_k) - v_{1,k} - v_{2,k} + w_{v,k}. \tag{8}$$

The measurement Jacobian is $H_k = [\partial U/\partial \mathrm{SOC},\ -1,\ -1,\ R_0]$: the bias is instantaneously observable through the ohmic term $R_0 b_k$ and, on longer horizons, through the coulombic channel $\partial \mathrm{SOC}_{k+1}/\partial b_k = \Delta t_k/(3600\,Q)$ weighted by the local OCV slope. Both channels degrade in flat OCV regions and for low-impedance cells, which is the mechanistic origin of the chemistry dependence investigated in the campaign. Analogously, a voltage-sensor-bias state $b_{v,k}$ can be appended (measurement Jacobian entry $\partial v/\partial b_v = 1$), either instead of or in addition to b_k; the filter with both bias states is included deliberately, as the simultaneous case is known to make the SOC unobservable in principle [9], and the campaign quantifies the practical consequences. The process noise of the bias state, q_b, sets the drift-tracking bandwidth of the filter, and the measurement variance is inflated above the true sensor noise ($r_v = 2.5 \times 10^{-5}\,\mathrm{V}^2$, i.e. an effective 5 mV standard deviation) to absorb the residual ECM model error and prevent it from being misattributed to the bias state. The observer can additionally (or alternatively) be augmented with a voltage-sensor bias state $b_{v,k}$, also modeled as a random walk, which enters the measurement model (8) additively with Jacobian entry +1. Estimating both b and b_v simultaneously is known to make the SOC weakly identifiable [9]; the campaign quantifies this cost empirically by comparing the $\{b\}$-, $\{b_v\}$- and $\{b, b_v\}$-augmented variants. The baseline observer is the identical EKF without any bias state; all filters are initialized with a deliberate SOC error to also assess convergence.

2.6 Battery ageing emulation

Battery ageing is emulated in the plant by two controlled mechanisms: capacity fade, implemented as a proportional loss of active material in both electrodes (the active-material volume fractions are scaled by the target state of health, SOH), and resistance growth, implemented as an additional series resistance ΔR applied to the terminal voltage, representing aged interfaces, welds and cabling. Five life stages are defined, jointly degrading the battery and the sensors: SOH $\in \{100, 95, 90, 85, 80\}\,\%$, $\Delta R \in \{0, 15, 30, 45, 60\}\,\%$ of the beginning-of-life (BOL) R_0, current-sensor offset $b_0 \in \{10, 40, 80, 110, 150\}\,\mathrm{mA}$ and voltage-sensor offset $b_v \in \{0.5, 1, 2, 3, 5\}\,\mathrm{mV}$. The true usable capacity of each aged cell is measured by a C/25 discharge of the aged plant, while the observer retains its BOL calibration (Q, ECM parameters and OCV)—the realistic situation of a BMS without online capacity adaptation. Note that with the BOL capacity in the coulomb-counting equation, capacity fade produces a state-equation error proportional to the current, whose running mean is absorbed by the bias state; the bias estimate therefore acquires the meaning of a *lumped ageing-and-sensor disturbance* rather than a pure sensor offset, which is quantified in the results.

2.7 Load profiles

Three load profiles are used. (i) A 1 C constant-current (CC) discharge from 100 % SOC to the voltage cut-off. (ii) A full CC–CV cycle: 1 C discharge, 10 min rest, 0.5 C charge with a 4.2 V

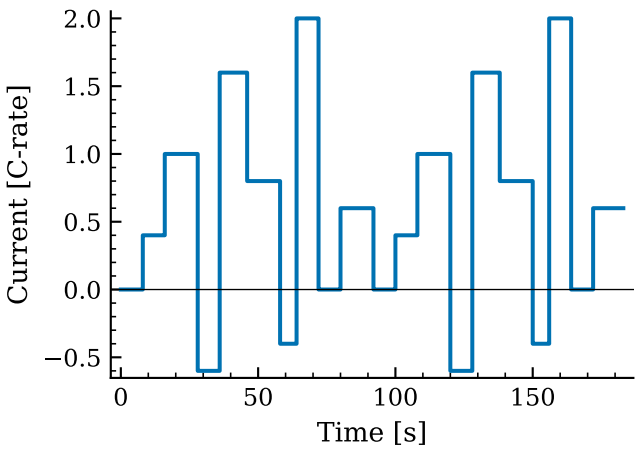


Figure 1: Two periods of the DST current profile, normalized to its peak discharge magnitude and scaled to a 2 C peak (positive = discharge).

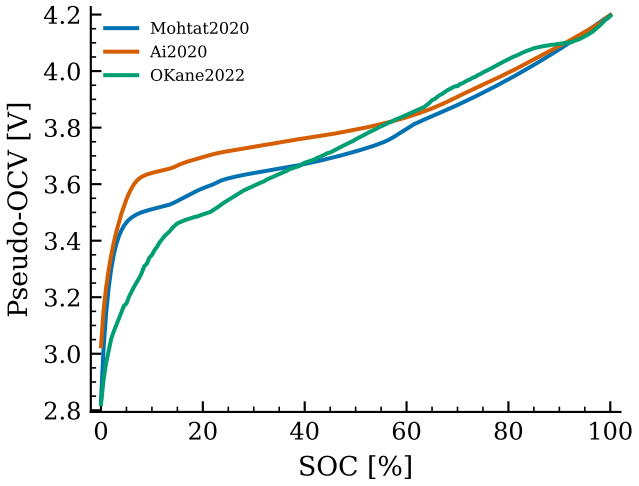


Figure 2: Pseudo-OCV curves of the three virtual cells (mean of C/25 charge and discharge branches, 25 °C).

constant-voltage hold to a C/50 cut, and a final rest. (iii) A dynamic stress test (DST) in the spirit of the USABC procedure [19]: the 92 s base sequence of discharge, rest and regenerative-charge steps is first normalized so that its peak discharge magnitude equals one, then scaled such that the peak reaches a prescribed C-rate, and repeated until a target depth of discharge (DoD) or the voltage cut-off is reached. The DST is applied after a CC–CV charge (0.5 C, 4.2 V, C/50 cut) so that each dynamic test starts from a well-defined full state. Figure 1 shows the scaled profile.

2.8 Simulation campaign and metrics

The campaign combines a baseline scenario (DST with 2 C peak, 80 % DoD, 25 °C, drift $\dot{b} = 0.15\,\mathrm{A/h}$ plus a 0.1 A mid-test step, $b_0 = 0$, $b_v = 0$) with one-factor-at-a-time sweeps: chemistry and profile (3 parameter sets × {CC, CC–CV, DST}), temperature (10 °C, 25 °C and 40 °C), DST peak C-rate (1–3C), DoD (50 % to 95 %), drift rate (0.05 A/h to 0.60 A/h), sensor age ($b_0 \in \{0, 0.2, 0.5\}$ A) and voltage bias ($b_v = 5\,\mathrm{mV}$, alone and combined with current drift), a datasheet-consistent *realistic* scenario ($b_0 = 50\,\mathrm{mA}$, $\dot{b} = 5\,\mathrm{mA/h}$, $b_v = 1\,\mathrm{mV}$), and the five-stage

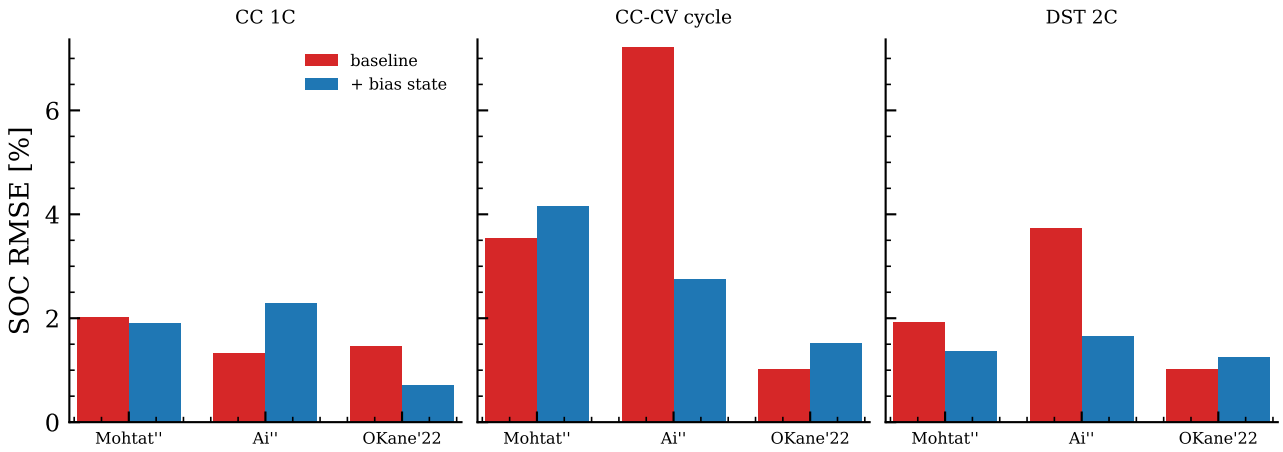


Figure 3: SOC RMSE of the baseline and bias-augmented EKF for the three chemistries under CC, CC–CV and DST operation (baseline bias scenario).

joint ageing campaign of Section 2.6, for 28 cases in total. Every case is run with both the baseline and the augmented EKF under identical tuning, noise realizations and (wrong) initialization. Performance is quantified by the SOC root-mean-square error (RMSE) and maximum absolute error after a 120 s transient, and by the RMSE of the bias estimate against the injected ground truth. All raw time series are stored, and all figures and tables in Section 3 are generated programmatically from them.

3 Results

[TODO: Results section intentionally left empty. The figures and the table below are auto-generated by `postprocess.py` from the campaign in `run_experiments.py`; write the discussion around them.]

3.1 Chemistry and load-profile dependence

3.2 Operating-condition sweeps

3.3 Drift tracking

3.4 Interaction with battery ageing

3.5 Interaction with battery aging

3.6 Current/voltage bias ambiguity

3.7 Experimental validation

[TODO: Add experimental validation on the laboratory tester: software-injected drifting bias with ground truth from a reference shunt, and a physical series resistance inserted in the cabling.]

4 Conclusions

[TODO: Write after results.]

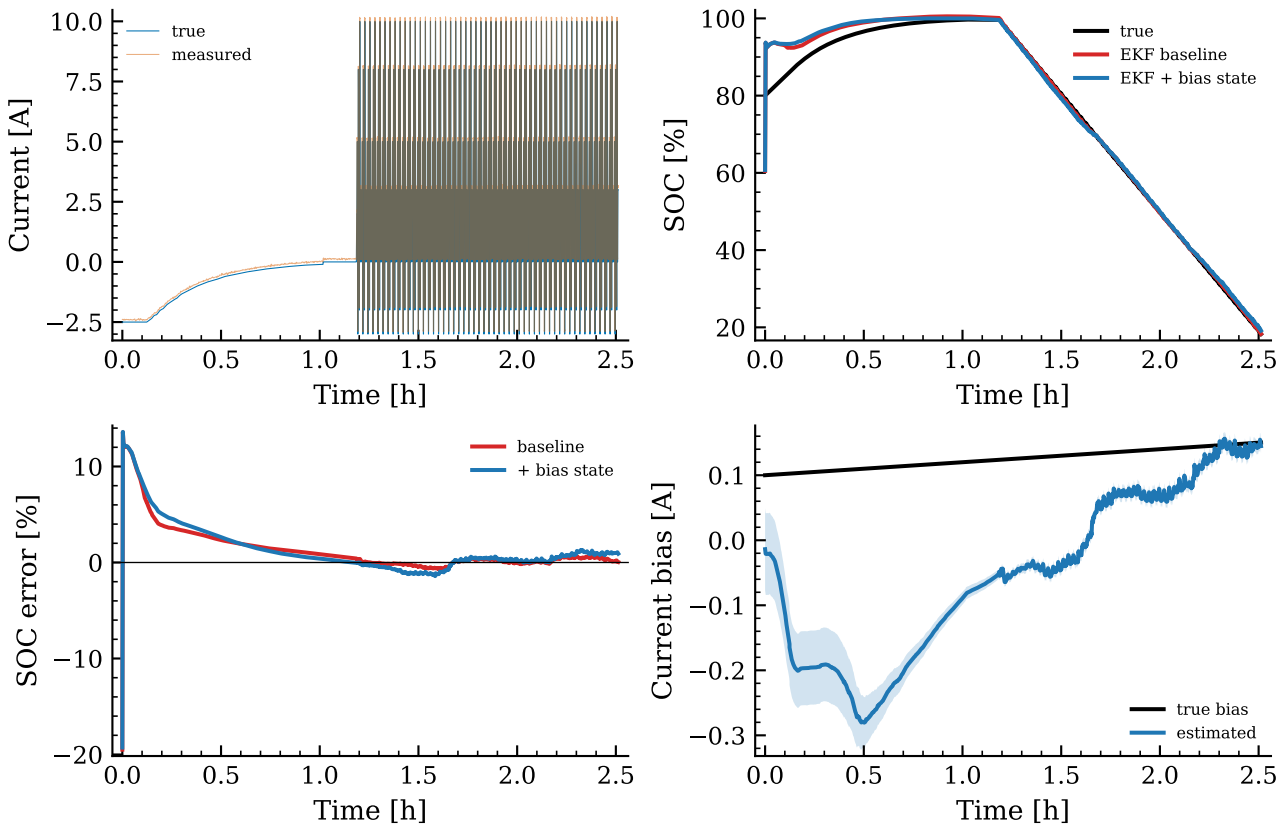


Figure 4: Representative DST case (OKane2022, 2 C peak, 25 °C): measured vs. true current, SOC estimates, SOC errors, and bias tracking with $\pm 2\sigma$ credible band.

Acknowledgments

[TODO: Funding and acknowledgments.]

References

[1] Xiaosong Hu, Shengbo Li, and Huei Peng. A comparative study of equivalent circuit models for Li-ion batteries. *Journal of Power Sources*, 198:359–367, 2012. doi: 10.1016/j.jpowsour.2011.10.013.

[2] Gregory L. Plett. Extended Kalman filtering for battery management systems of LiPB-based HEV battery packs: Part 1. background. *Journal of Power Sources*, 134(2):252–261, 2004. doi: 10.1016/j.jpowsour.2004.02.031.

[3] Gregory L. Plett. Extended Kalman filtering for battery management systems of LiPB-based HEV battery packs: Part 2. modeling and identification. *Journal of Power Sources*, 134(2):262–276, 2004. doi: 10.1016/j.jpowsour.2004.02.032.

[4] Gregory L. Plett. Extended Kalman filtering for battery management systems of LiPB-based HEV battery packs: Part 3. state and parameter estimation. *Journal of Power Sources*, 134 (2):277–292, 2004. doi: 10.1016/j.jpowsour.2004.02.033.

[5] Jiangtao He, Shujuan Meng, and Fengjun Yan. A comparative study of SOC estimation based on equivalent circuit models. *Frontiers in Energy Research*, 10:914291, 2022. doi: 10.3389/fenrg.2022.914291.

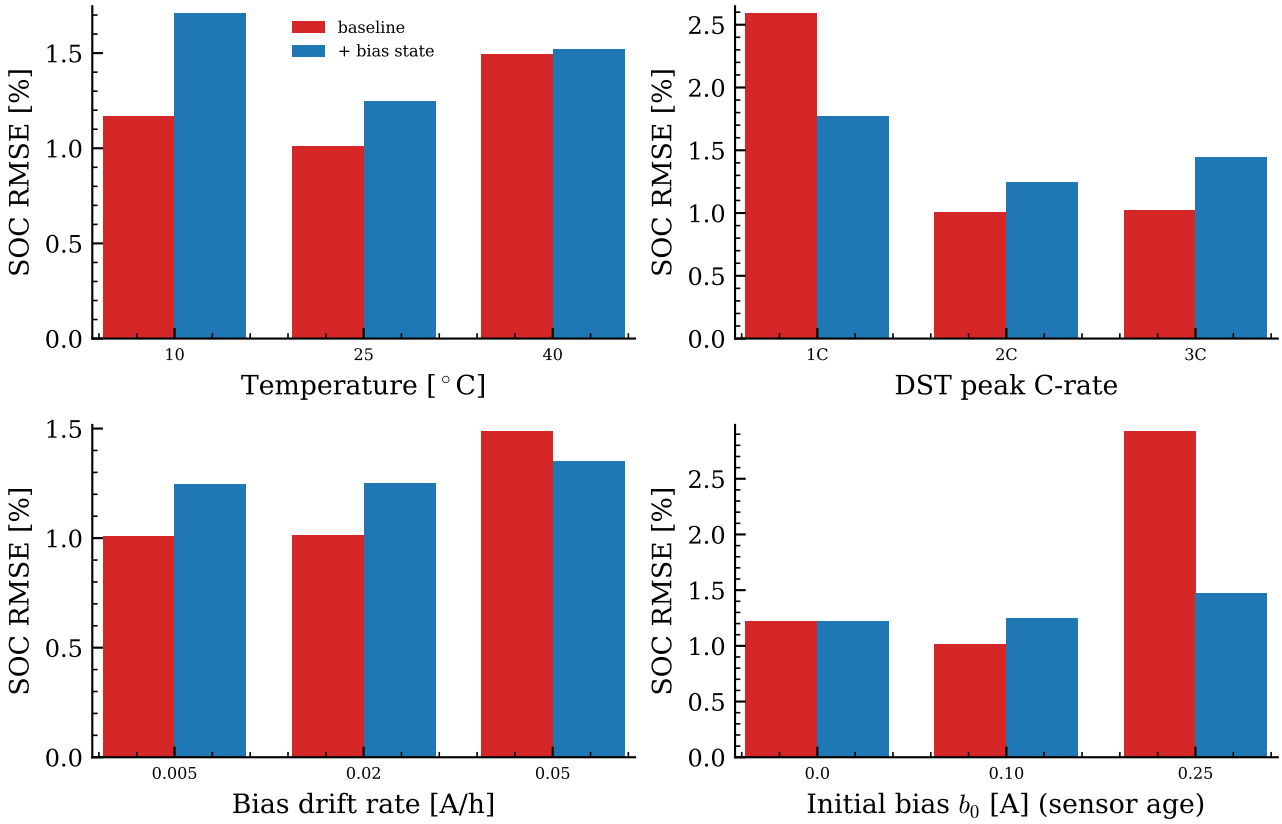


Figure 5: One-factor-at-a-time sweeps (OKane2022, DST): temperature, DST peak C-rate, bias drift rate and sensor age.

[6] Yuejiu Zheng, Minggao Ouyang, Xuebing Han, Languang Lu, and Jianqiu Li. Investigating the error sources of the online state of charge estimation methods for lithium-ion batteries in electric vehicles. *Journal of Power Sources*, 377:161–188, 2018. doi: 10.1016/j.jpowsour.2017.11.094. VERIFY volume/pages against published version.

[7] Cyril Truchot, Matthieu Dubarry, and Bor Yann Liaw. State-of-charge estimation and uncertainty for lithium-ion battery strings. *Applied Energy*, 119:218–227, 2014. doi: 10.1016/j.apenergy.2013.12.046. VERIFY volume/pages against published version.

[8] Xinfan Lin. Theoretical analysis of battery SOC estimation errors under sensor bias and variance. *IEEE Transactions on Industrial Electronics*, 65(9):7138–7148, 2018. doi: 10.1109/TIE.2018.2795521. VERIFY issue number and DOI against published version.

[9] Shi Zhao, Stephen R. Duncan, and David A. Howey. Observability analysis and state estimation of lithium-ion batteries in the presence of sensor biases. *IEEE Transactions on Control Systems Technology*, 25(1):326–333, 2017. doi: 10.1109/TCST.2016.2542115. Authors verified; VERIFY volume/pages against published version.

[10] Min Young Yoo, Jae Hyun Lee, Joo-Ho Choi, Jin Sul Huh, and Woosuk Sung. State-of-charge estimation of batteries for hybrid urban air mobility. *Aerospace*, 10(6):550, 2023. doi: 10.3390/aerospace10060550. VERIFY author given names.

[11] Feng Guo, Luis D. Couto, Khiem Trad, Guangdi Hu, and Mohammadhosein Safari. Residual bias compensation filter for physics-based SOC estimation in lithium iron phosphate batteries, 2025. Submitted to the European Control Conference (ECC) 2026.

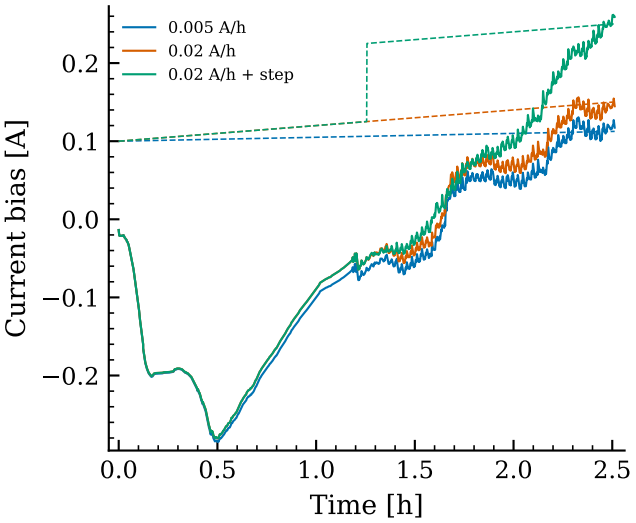


Figure 6: Estimated (solid) vs. true (dashed) current-sensor bias for three drift rates.

[12] Tianyu Lan, Zhi-Wei Gao, Haishuang Yin, and Yuanhong Liu. A sensor-fault-estimation method for lithium-ion batteries in electric vehicles. *Sensors*, 23(18):7737, 2023. doi: 10.3390/s23187737.

[13] Valentin Sulzer, Scott G. Marquis, Robert Timms, Martin Robinson, and S. Jon Chapman. Python battery mathematical modelling (PyBaMM). *Journal of Open Research Software*, 9(1):14, 2021. doi: 10.5334/jors.309.

[14] Peyman Mohtat, Suhak Lee, Valentin Sulzer, Jason B. Siegel, and Anna G. Stefanopoulou. Differential expansion and voltage model for li-ion batteries at practically relevant temperatures. *Journal of The Electrochemical Society*, 167(11):110561, 2020. doi: 10.1149/1945-7111/aba5d1. VERIFY author list against published version.

[15] Weilong Ai, Ludwig Kraft, Johannes Sturm, Andreas Jossen, and Billy Wu. Electrochemical thermal-mechanical modelling of stress inhomogeneity in lithium-ion pouch cells. *Journal of The Electrochemical Society*, 167(1):013512, 2020. doi: 10.1149/2.0122001JES.

[16] Simon E. J. O'Kane, Weilong Ai, Ganesh Madabattula, Diego Alonso-Alvarez, Robert Timms, Valentin Sulzer, Jacqueline S. Edge, Billy Wu, Gregory J. Offer, and Monica Marinescu. Lithium-ion battery degradation: how to model it. *Physical Chemistry Chemical Physics*, 24:7909–7922, 2022. doi: 10.1039/D2CP00417H.

[17] Chang-Hui Chen, Ferran Brosa Planella, Kieran O'Regan, Dominika Gastol, W. Dhammika Widanage, and Emma Kendrick. Development of experimental techniques for parameterization of multi-scale lithium-ion battery models. *Journal of The Electrochemical Society*, 167(8):080534, 2020. doi: 10.1149/1945-7111/ab9050.

[18] Melexis. A view to a battery current sensor: shunt or Hall? Technical whitepaper, `https://www.melexis.com/en/news/tech-talks/a-view-to-a-battery-current-sensor-shunt-or-hall`. Accessed 2026-07-01. VERIFY publication year; consider replacing with an archival source on sensor drift.

[19] USABC. Electric vehicle battery test procedures manual, revision 2. Technical report, United States Advanced Battery Consortium / Idaho National Engineering Laboratory, 1996.

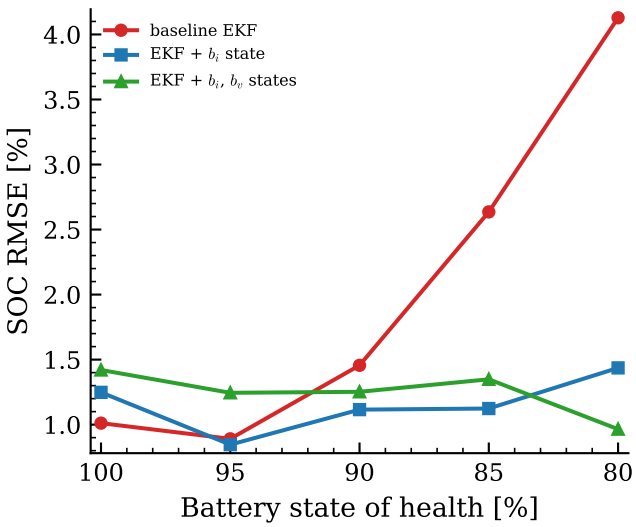


Figure 7: Joint battery-ageing × sensor-ageing campaign (OKane2022, DST 2 C peak): SOC RMSE of the baseline EKF and of the variants augmented with the current-sensor bias state and with both bias states, as a function of battery state of health. The observer retains its beginning-of-life calibration at all stages.

[20] Scott G. Marquis, Valentin Sulzer, Robert Timms, Colin P. Please, and S. Jon Chapman. An asymptotic derivation of a single particle model with electrolyte. *Journal of The Electrochemical Society*, 166(15):A3693–A3706, 2019. doi: 10.1149/2.0341915jes.

[21] Marc Doyle, Thomas F. Fuller, and John Newman. Modeling of galvanostatic charge and discharge of the lithium/polymer/insertion cell. *Journal of The Electrochemical Society*, 140 (6):1526–1533, 1993. doi: 10.1149/1.2221597.

[22] Allegro MicroSystems. Non-intrusive hall-effect current-sensing techniques for power electronics. Technical document, Allegro MicroSystems. Available at `https://www.allegromicro.com`; accessed 2026.

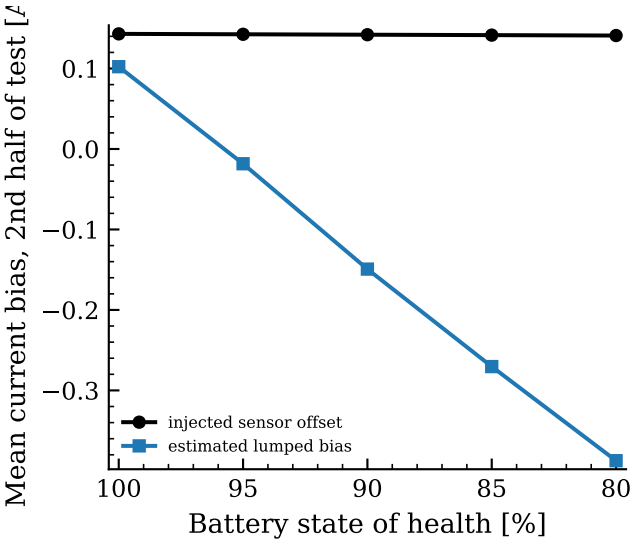


Figure 8: Mean estimated lumped bias over the second half of each test vs. the injected physical sensor offset, across life stages: the bias state increasingly absorbs ageing-induced disturbances (capacity fade, resistance growth) on top of the sensor offset.

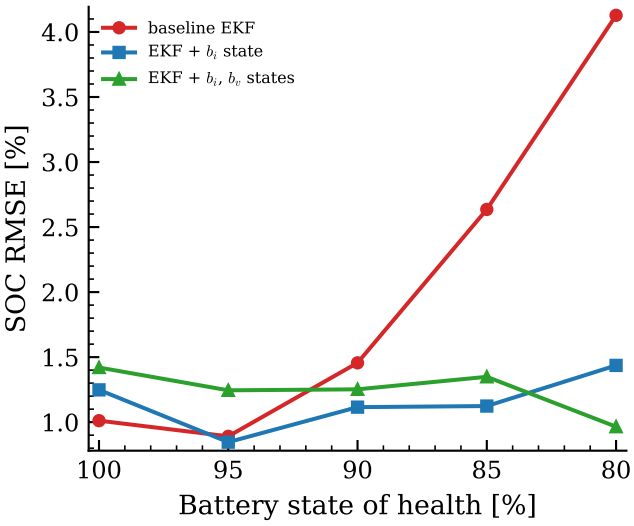


Figure 9: SOC RMSE vs. battery SOH for the baseline, current-bias augmented, and dual-bias EKF; observer calibration is kept at the fresh state throughout.

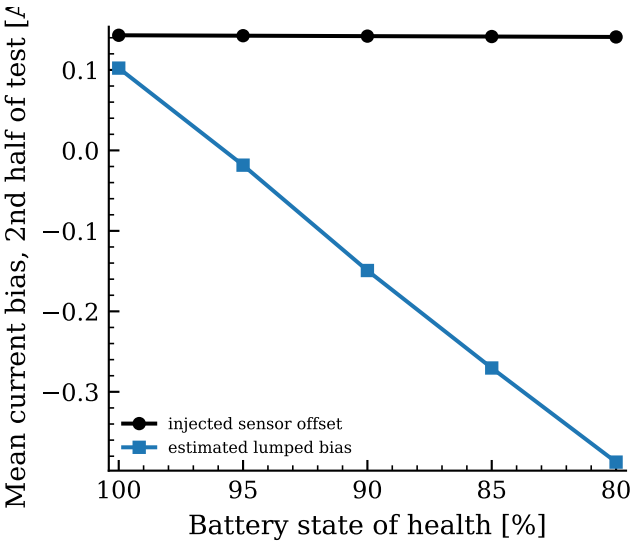


Figure 10: Injected sensor offset vs. estimated lumped bias across SOH levels: the bias state absorbs aging-induced disturbances beyond the true sensor offset (diagnosis/accuracy confounding).

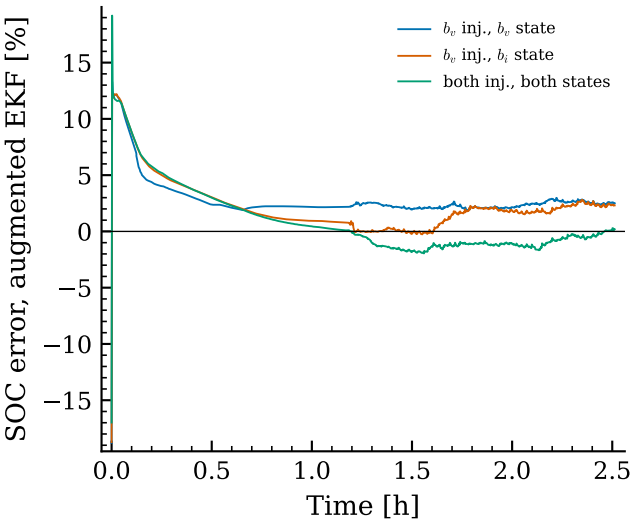


Figure 11: SOC error of the augmented EKF when a 5 mV voltage-sensor bias is present, alone and combined with current drift.

Table 1: Summary of all simulation cases. RMSE values computed after a 120 s transient. Baseline: EKF without bias state; augmented: EKF with current-bias state.

Case	Profile	T [°C]	drift [A/h]	b_0 [A]	b_v [mV]	$\mathrm{RMSE}_{\mathrm{base}}$ [%]	$\mathrm{RMSE}_{\mathrm{aug}}$ [%]
Mohtat2020_cc	cc	25	0.02	0.1	0	2.01	1.90
Mohtat2020_cccv	cccv	25	0.02	0.1	0	3.54	4.15
Mohtat2020_dst	dst	25	0.02	0.1	0	1.92	1.36
Ai2020_cc	cc	25	0.02	0.1	0	1.32	2.28
Ai2020_cccv	cccv	25	0.02	0.1	0	7.22	2.74
Ai2020_dst	dst	25	0.02	0.1	0	3.74	1.64
OKane2022_cc	cc	25	0.02	0.1	0	1.45	0.71
OKane2022_cccv	cccv	25	0.02	0.1	0	1.01	1.52
OKane2022_dst	dst	25	0.02	0.1	0	1.01	1.25
OKane2022_dst_T10	dst	10	0.02	0.1	0	1.17	1.71
OKane2022_dst_T40	dst	40	0.02	0.1	0	1.50	1.52
OKane2022_dst_peak1C	dst	25	0.02	0.1	0	2.60	1.77
OKane2022_dst_peak3C	dst	25	0.02	0.1	0	1.02	1.45
OKane2022_dst_dod50	dst	25	0.02	0.1	0	0.95	0.98
OKane2022_dst_dod95	dst	25	0.02	0.1	0	1.01	0.87
OKane2022_dst_drift0p005	dst	25	0.01	0.1	0	1.01	1.25
OKane2022_dst_drift0p05	dst	25	0.05	0.1	0	1.49	1.35
OKane2022_dst_stepfault	dst	25	0.02	0.1	0	1.78	1.45
OKane2022_dst_age0p0	dst	25	0.02	0.0	0	1.22	1.23
OKane2022_dst_age0p25	dst	25	0.02	0.2	0	2.92	1.47
OKane2022_dst_vbias5mV_obsV	dst	25	0.00	0.0	5	2.23	2.49
OKane2022_dst_vbias5mV_obsI	dst	25	0.00	0.0	5	2.23	1.92
OKane2022_dst_bothbias_obsIV	dst	25	0.02	0.1	5	0.81	1.56
OKane2022_dst_soh100	dst	25	0.02	0.1	0	1.01	1.25
OKane2022_dst_soh95	dst	25	0.02	0.1	0	0.89	0.85
OKane2022_dst_soh90	dst	25	0.02	0.1	0	1.46	1.12
OKane2022_dst_soh85	dst	25	0.02	0.1	0	2.64	1.12
OKane2022_dst_soh80	dst	25	0.02	0.1	0	4.13	1.44